

Generalized State-Space Averaging Modeling of Dual-Active-Bridge Converter With Triple-Phase-Shift Control

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Abstract—This paper proposes a generalized model based on the generalized state-space averaging model for dual-active-bridge (DAB) DC-DC converters. The proposed model is suitable for all modulation modes, and the model is general, and the model will not be different due to the change of the transmission power mode. The model uses the AC first-order component of the inductor current and the DC component of the capacitor voltage as state variables to establish the state equation. Based on the large-signal model, the small-signal model and the transfer function from the control variable to the output voltage are derived for the DAB converter. Finally, an experimental prototype of DAB converter is built, and small-signal analysis is carried out based on the simulation software and the experimental prototype. Comparing the proposed model with the reduced-order model, the proposed model is more matched with the experimental and simulation results than the reduced-order model, which verifies the accuracy of the model.

Keywords—DAB converter, generalized state-space averaging method, modeling, triple-phase-shift control

I. INTRODUCTION

Due to its high power density, high reliability, high transmission power, large power capacity and soft switching[1], dual active bridge converters are popular and widely used in electric vehicles, photovoltaic power generation, etc., with bidirectional energy flow in the energy storage system.

According to the different number of control degrees of freedom, it can be divided into 4 modulation modes. Among them, single-phase-shift (SPS) modulation has only one control degree of freedom, extended-phase-shift (EPS) modulation[2] and double-phase-shift (DPS) modulation have two control degrees of freedom, and triple-phase-shift (TPS) modulation[3] has three independent control degrees of freedom. Other modulation methods can be regarded as a special case of TPS modulation. More control degrees of freedom can realize the global optimal control of the converter, such as current stress optimization, reflow power optimization, soft switching optimization and efficiency optimization, etc., so it is very important to establish a DAB converter model with TPS control.

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In the DAB converter, since the inductor current is a pure AC component, the small ripple assumption in the traditional state-space averaging method modeling is not satisfied. Therefore, the traditional state-space averaging method is not suitable for DAB converter. There are three common modeling approaches for DAB converters. One is a reduced-order model [4]. This method uses the average current of input and output to build a model, and its calculation is simple, but the accuracy of the model is low due to ignoring the dynamic characteristics of the inductor current. On this basis, the input and output filters are considered to make the model closer to the actual situation, but the computational complexity is increased, and the model is only suitable for SPS control [5]. In [6], a more complete power loss model is considered, including input and output filters, conduction, inductors, and transformers. Although this model is suitable for TPS control, the calculation is complicated, and the model needs to be classified and discussed according to the difference among the three control variables. The second model is the discrete time model[7]-[8], which calculates the state equation in each time interval in a switching cycle, and can obtain an accurate solution of the inductor current, and the model accuracy is high. But this model is only suitable for SPS control, and the calculation is complicated. If it is to be extended to TPS control, due to the influence of three control degrees of freedom, there are many states in one switching cycle and it is difficult to derive. The third is the generalized state-space averaging method, and it is a more general averaging method that is theoretically suitable for arbitrary waveforms. The generalized state-space average model considers the dynamic characteristics of the inductor current, and the model has high accuracy. In [9], a generalized state space average model is built, but the model is only applicable to the case of SPS control. Subsequently, the model is extended to the case of DPS control [10]. However, no verification of the small-signal model was performed.

Aiming at the low accuracy of the DAB converter model and the problem that the model is only based on SPS modulation, this paper adopts the generalized state average method to build the DAB converter model under TPS control. First, the inductor current and capacitor voltage of the DAB converter are selected as state variables to establish the state equation. And then the Fourier series is used to represent each state variable, and the coefficients of the Fourier series are

used to construct a large-signal model. Second, small disturbances are added at the steady-state operating point, and the small-signal model and the transfer function of the control variable to the output voltage are derived. Finally, the correctness of the model is verified by simulation and experiment. Compared with the reduced-order model, the proposed model has higher accuracy and the model does not need to be classified and discussed, and is not affected by the operation condition of the converter, which verifies the correctness of the theoretical analysis.

II. DAB CONVERTER WITH TPS CONTROL

As shown in Fig. 1 the DAB converter consists of two H bridge cells and a high-frequency transformer (HFT), where the transformer ratio is $n:1$. Bridge H₁ of primary side inverter, bridge H₂ of secondary side rectifier, and u_1, u_2 is the ac terminal voltage of H₁, H₂, respectively. L is the summation of the leakage inductance of the HFT and auxiliary inductor, and i_L is the current of L . R_s represents the sum of transistor on-resistance and transformer winding resistance. R represents the load resistance. C represents the output capacitance. V_1 represents the input voltage, and v_2 represents the output voltage.

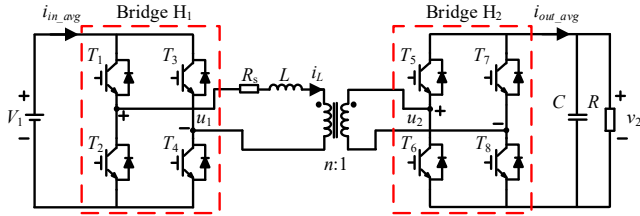


Fig. 1. DAB converter schematic.

Using TPS, the converter has three control variables d_1, d_2, d_3 , where $0 \leq d_1 \leq 1, -1 \leq d_2 \leq 1, 0 \leq d_3 \leq 1$. When $d_2 > 0$, power is transmitted from bridge H₁ to bridge H₂, and when $d_2 < 0$, power is transmitted from bridge H₂ to bridge H₁. $d_1 \cdot T_{hs}$ represents the phase shift angle between the switches T_1 and T_4 in bridge H₁, $d_2 \cdot T_{hs}$ represents the phase shift angle between the switches T_1 and T_4 . $d_3 \cdot T_{hs}$ represents the phase shift angle between the switches T_5 and T_8 in bridge H₂, where T_{hs} is half switching period T , T equals to $1/f_s$, and f_s is the switching frequency. Fig. 2 depicts the main operating waveforms of the DAB converter.

III. REDUCED-ORDER MODEL OF DAB CONVERTER

The reduced-order model is a simplified model that ignores the inductor current dynamics. The input and output average currents are solved by the transmission power expression, and then the large-signal model and the small-signal model of the DAB converter are constructed by using the input and output average currents.

With the TPS control, due to the influence of the three control variables d_1, d_2, d_3 , the transmission power modes can be divided into 12 kinds[11]. This paper takes one of the transmission power modes as an example to build the reduced-order model of the DAB converter. The transmission power P of DAB converter with TPS control is expressed as (1), where $0 \leq d_1 \leq d_2, 0 \leq d_2 \leq 1 - d_2, 0 \leq d_3 \leq 1$.

$$P = \frac{nv_1v_2}{4f_sL} [d_1(-d_1 + 2d_2 + d_3 - 1) - 2d_2(d_2 + d_3 - 1) + d_3(1 - d_3)] \quad (1)$$

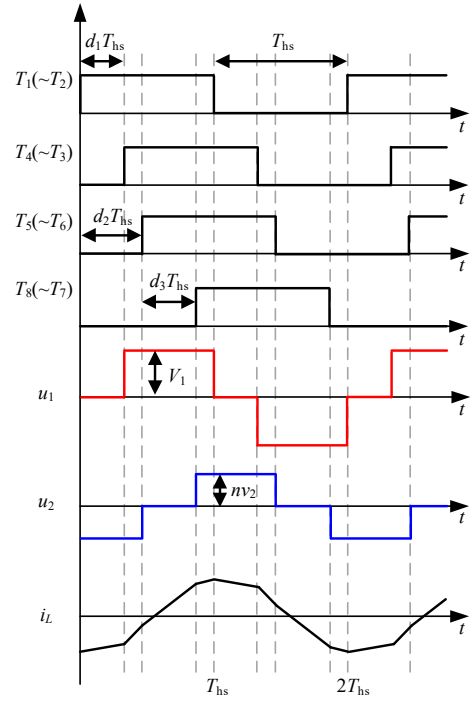


Fig. 2. Operation of DAB converter with TPS control.

The average value of the input and output currents is obtained by (2) and (3). Therefore, the large-signal model of the DAB reduced-order model can be expressed as Fig. 3.

$$i_{in_avg} = \frac{nv_2}{4f_sL} [d_1(-d_1 + 2d_2 + d_3 - 1) - 2d_2(d_2 + d_3 - 1) + d_3(1 - d_3)] \quad (2)$$

$$i_{out_avg} = \frac{nv_1}{4f_sL} [d_1(-d_1 + 2d_2 + d_3 - 1) - 2d_2(d_2 + d_3 - 1) + d_3(1 - d_3)] \quad (3)$$

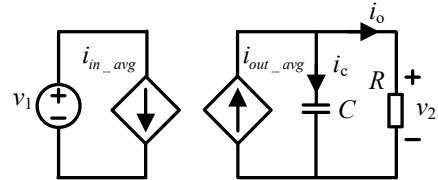


Fig. 3. Large-signal model of the DAB reduced-order model.

By adding small disturbances at the control variables d_1, d_2, d_3, v_1 and v_2 , we can obtain the small-signal equations of the input and output currents and obtain the small-signal model of the DAB reduced-order model as shown in Figure 4.

$$\hat{i}_{in_avg} = k_1\hat{v}_2 + k_2\hat{d}_1 + k_3\hat{d}_2 + k_4\hat{d}_3 \quad (4)$$

$$\hat{i}_{out_avg} = k_5\hat{v}_1 + k_6\hat{d}_1 + k_7\hat{d}_2 + k_8\hat{d}_3 \quad (5)$$

where

$$k_1 = k_5 = \frac{n}{4f_sL} [D_1(-D_1 + 2D_2 + D_3 - 1) - 2D_2(D_2 + D_3 - 1) + D_3(1 - D_3)]$$

$$k_2 = k_6 = \frac{nV_1}{4f_sL} (-2D_1 + 2D_2 + D_3 - 1)$$

$$k_3 = k_7 = \frac{nV_1}{4f_sL} (2D_1 - 4D_2 - 2D_3 + 2)$$

$$k_4 = k_8 = \frac{nV_1}{4f_s L} (D_1 - 2D_2 - 2D_3 + 1)$$

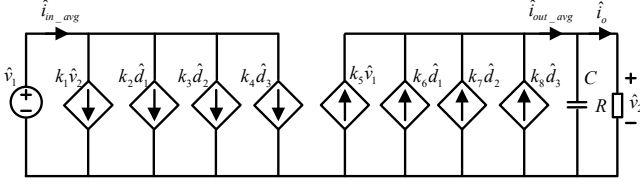


Fig. 4. Small-signal model of the DAB reduced-order model.

Based on Fig. 3 and (5), and consider $\hat{v}_1=0$, the control block diagram from the control signal d_1 , d_2 and d_3 to the output voltage v_2 is shown in Fig. 5. The transfer function of control variable d_1 , d_2 and d_3 to the output voltage v_2 are (6), (7) and (8).

$$G_{\hat{v}_2 \hat{d}_1}(s) = k_6 \cdot \frac{R}{sRC + 1} \quad (6)$$

$$G_{\hat{v}_2 \hat{d}_2}(s) = k_7 \cdot \frac{R}{sRC + 1} \quad (7)$$

$$G_{\hat{v}_2 \hat{d}_3}(s) = k_8 \cdot \frac{R}{sRC + 1} \quad (8)$$

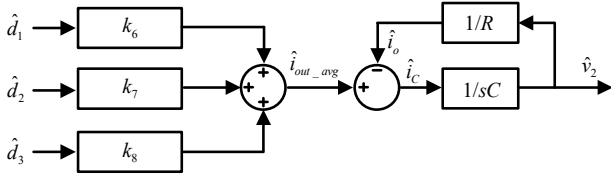


Fig. 5. Control block diagram from control to output.

In this section, we can obtain the reduced-order model of the DAB converter with TPS control, but this model represents only one of the 12 transmission power modes, and the model will no longer be applicable if the transmission power mode is changed. Therefore, the reduced-order model is not universal and cannot apply to all the power modes. Besides, since the dynamic characteristics of the inductor current are ignored, the equivalent resistance R_s of the transistor on-time resistance and transformer winding resistance cannot be considered, and the model accuracy is low.

IV. GENERALIZED MODEL OF DAB CONVERTER

A. Large-Signal Model of DAB

In Fig. 2, it can be seen from the figure that u_1 and u_2 has three different states in one switching cycle. Therefore, $u_1(\tau)=S_1(\tau) \cdot v_1(\tau)$, $u_2(\tau)=S_2(\tau) \cdot v_2(\tau)$. where the switching function $S_1(\tau)$ and $S_2(\tau)$ are

$$S_1 = \begin{cases} 1 & d_1 T_{hs} \leq \tau < T_{hs} \\ 0 & 0 \leq \tau < d_1 T_{hs}, \\ & T_{hs} \leq \tau < (1+d_1)T_{hs} \\ -1 & (1+d_1)T_{hs} \leq \tau < 2T_{hs} \end{cases} \quad (9)$$

$$S_2 = \begin{cases} 1 & (d_2+d_3)T_{hs} \leq \tau < (1+d_2)T_{hs} \\ 0 & d_2 T_{hs} \leq \tau < (d_2+d_3)T_{hs}, \\ & (1+d_2)T_{hs} \leq \tau < (1+d_2+d_3)T_{hs} \\ -1 & 0 \leq \tau < d_2 T_{hs}, \\ & (1+d_2+d_3)T_{hs} \leq \tau < 2T_{hs} \end{cases} \quad (10)$$

Select the inductor currents i_L and the capacitor voltages v_2 as state variables to establish the DAB state equations, expressed as

$$\frac{di_L(\tau)}{d\tau} = -\frac{R_s}{L} i_L(\tau) + \frac{1}{L} S_1(\tau) \cdot V_1 - \frac{n}{L} S_2(\tau) \cdot v_2(\tau) \quad (11)$$

$$\frac{dv_2(\tau)}{d\tau} = \frac{n}{C} S_2(\tau) \cdot i_L(\tau) - \frac{1}{RC} v_2(\tau) \quad (12)$$

The key to the generalized state space averaging method is to use the Fourier series instead of the average state variable. The Fourier series of the state variable $x(\tau)$ in an interval $t-T < \tau < t$ is expressed as

$$x(\tau) = \sum_{k=-\infty}^{\infty} \langle x \rangle_k(t) e^{jk\omega_s \tau} \quad (13)$$

where $\omega_s = 2\pi f_s$, $\langle x \rangle_k(t)$ represents the k th coefficient of the Fourier series, which can be calculated by (14).

$$\langle x \rangle_k(t) = \frac{1}{T} \int_{t-T}^t x(\tau) e^{-jk\omega_s \tau} d\tau \quad (14)$$

The two properties of differentiation and convolution of the Fourier series are applied[12], which are expressed as

$$\frac{d}{dt} \langle x \rangle_k(t) = \left\langle \frac{dx}{dt} \right\rangle_k(t) - jk\omega_s \langle x \rangle_k(t) \quad (15)$$

$$\langle xy \rangle_k(t) = \sum_{i=-\infty}^{\infty} \langle x \rangle_{k-i}(t) \langle y \rangle_i(t) \quad (16)$$

For the DAB converter in Fig. 1(a), the switching state variables $S_1(\tau)$, $S_2(\tau)$, and state variable i_L are seen as pure AC components, whereas the state variable v_2 is seen as pure DC component. To simplify the model, only the 0th and 1st order components of state variables are considered in this paper. Applying(15)-(16) to the state-space equations in (11) and (12). The state equation is reconstructed by the coefficients of the Fourier series.

$$\frac{d\langle i_L \rangle_{1R}}{d\tau} = \omega_s \langle i_L \rangle_{1I} - \frac{R_s}{L} \langle i_L \rangle_{1R} + \frac{1}{L} \langle S_1 \rangle_{1R} \cdot V_1 - \frac{n}{L} \langle S_2 \rangle_{1R} \cdot \langle v_2 \rangle_0 \quad (17)$$

$$\frac{d\langle i_L \rangle_{1I}}{d\tau} = -\omega_s \langle i_L \rangle_{1R} - \frac{R_s}{L} \langle i_L \rangle_{1I} + \frac{1}{L} \langle S_1 \rangle_{1I} \cdot V_1 - \frac{n}{L} \langle S_2 \rangle_{1I} \cdot \langle v_2 \rangle_0 \quad (18)$$

$$\frac{d\langle v_2 \rangle_0}{d\tau} = \frac{n}{C} (\langle S_2 \rangle_{1I} \cdot \langle i_L \rangle_{-1} + \langle S_2 \rangle_{-1I} \cdot \langle i_L \rangle_{1I}) - \frac{1}{RC} \langle v_2 \rangle_0 \quad (19)$$

where the subscript 'R' and 'I' represent the real, imaginary part of the 1st coefficients, respectively.

Similarly, the 1st order coefficients of the Fourier series of the switching functions $S_1(\tau)$ and $S_2(\tau)$ are

$$\langle s_1 \rangle_{1R} = -\frac{\sin(d_1\pi)}{\pi} \quad (20)$$

$$\langle s_1 \rangle_{1I} = -\frac{\cos(d_1\pi)+1}{\pi} \quad (21)$$

$$\langle s_2 \rangle_{1R} = -\frac{\sin(d_2\pi)+\sin(d_2\pi+d_3\pi)}{\pi} \quad (22)$$

$$\langle s_2 \rangle_{1I} = -\frac{\cos d_2\pi + \cos(d_2\pi + d_3\pi)}{\pi} \quad (23)$$

Finally, applying (20)-(23) into (17)-(19), and organize differential equations into matrix representation in (24).

$$\frac{d}{dt} \begin{bmatrix} \langle i_L \rangle_{1R} \\ \langle i_L \rangle_{1I} \\ \langle v_2 \rangle_0 \end{bmatrix} = \begin{bmatrix} -\frac{R_s}{L} & \omega_s & \frac{n}{\pi L} y \\ -\omega_s & -\frac{R_s}{L} & \frac{n}{\pi L} m \\ -\frac{2n}{\pi C} y & -\frac{2n}{\pi C} m & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} \langle i_L \rangle_{1R} \\ \langle i_L \rangle_{1I} \\ \langle v_2 \rangle_0 \end{bmatrix} + \begin{bmatrix} -\frac{\sin(d_1\pi)}{\pi L} \\ -\frac{\cos(d_1\pi)+1}{\pi L} \\ 0 \end{bmatrix} V_1 \quad (24)$$

The variables y and m are represented as

$$y = \sin d_2\pi + \sin(d_2\pi + d_3\pi) \quad (25)$$

$$m = \cos d_2\pi + \cos(d_2\pi + d_3\pi) \quad (26)$$

As analyzed above, we can obtain the large-signal model of the DAB converter as shown in (24). Set the left side of the equation equal to 0 to solve for the steady-state solution for the variables $\langle i_L \rangle_{1R}$, $\langle i_L \rangle_{1I}$ and $\langle v_2 \rangle_0$.

B. Small-Signal Model of DAB

By adding a small disturbance at the steady-state operating point, the small signal model of the DAB converter can be obtained. The small disturbance is defined as follows

$$\begin{cases} d_1 = D_1 + \hat{d}_1 \\ d_2 = D_2 + \hat{d}_2 \\ d_3 = D_3 + \hat{d}_3 \\ v_{20} = V_{20} + \hat{v}_{20} \\ \langle i_L \rangle_{1R} = I_{L1R} + \hat{i}_{L1R} \\ \langle i_L \rangle_{1I} = I_{L1I} + \hat{i}_{L1I} \end{cases} \quad (27)$$

Apply (27) into (24), there are non-linear components $\sin x$ and $\cos x$ in the formula, and use (28)-(29) to linearize the non-linear components[13].

$$\sin(D\pi + \hat{d}\pi) = \sin D\pi + \cos D\pi \cdot \hat{d}\pi \quad (28)$$

$$\cos(D\pi + \hat{d}\pi) = \cos D\pi - \sin D\pi \cdot \hat{d}\pi \quad (29)$$

Combining equations (24) and (27)-(29), eliminating the steady-state components and high-order components and simplifying the equations into a matrix form, the small signal model of the DAB converter can be obtained as shown in (30)

$$\frac{d}{dt} \begin{bmatrix} \hat{i}_{L1R} \\ \hat{i}_{L1I} \\ \hat{v}_{20} \end{bmatrix} = \begin{bmatrix} -\frac{R_s}{L} & \omega_s & \frac{n}{\pi L} Y \\ -\omega_s & -\frac{R_s}{L} & \frac{n}{\pi L} M \\ -\frac{2n}{\pi C} Y & -\frac{2n}{\pi C} M & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} \hat{i}_{L1R} \\ \hat{i}_{L1I} \\ \hat{v}_{20} \end{bmatrix} + \begin{bmatrix} -\frac{\cos(D_1\pi)V_1}{L} & \frac{nV_{20}}{L} M & \frac{nV_{20}}{L} M_1 \\ \frac{\sin(D_1\pi)V_1}{L} & -\frac{nV_{20}}{L} Y & -\frac{nV_{20}}{L} Y_1 \\ 0 & -\frac{2n}{C_o}(I_{L1R}M - I_{L1I}Y) & -\frac{2n}{C_o}(I_{L1R}M_1 - I_{L1I}Y_1) \end{bmatrix} \begin{bmatrix} \hat{d}_1 \\ \hat{d}_2 \\ \hat{d}_3 \end{bmatrix} \quad (30)$$

where

$$Y = \sin D_2\pi + \sin(D_2\pi + D_3\pi) \quad (31)$$

$$M = \cos D_2\pi + \cos(D_2\pi + D_3\pi) \quad (32)$$

$$Y_1 = \sin(D_2\pi + D_3\pi) \quad (33)$$

$$M_1 = \cos(D_2\pi + D_3\pi) \quad (34)$$

The transfer function of each control variable to the output voltage can be expressed as

$$G_{\hat{v}_2\hat{d}_1}(s) \Big|_{\hat{d}_2=0, \hat{d}_3=0} = C(sI - A)^{-1} B_1 \quad (35)$$

$$G_{\hat{v}_2\hat{d}_2}(s) \Big|_{\hat{d}_1=0, \hat{d}_3=0} = C(sI - A)^{-1} B_2 \quad (36)$$

$$G_{\hat{v}_2\hat{d}_3}(s) \Big|_{\hat{d}_1=0, \hat{d}_2=0} = C(sI - A)^{-1} B_3 \quad (37)$$

where

$$C = [0 \quad 0 \quad 1]$$

$$B_1 = \begin{bmatrix} -\frac{\cos(D_1\pi)V_1}{L} \\ \frac{\sin(D_1\pi)V_1}{L} \\ 0 \end{bmatrix}$$

$$B_2 = \begin{bmatrix} \frac{nV_{20}}{L} Z \\ -\frac{nV_{20}}{L} Y \\ -\frac{2n}{C_o}(I_{L1R}Z - I_{L1I}Y) \end{bmatrix}$$

$$B_3 = \begin{bmatrix} \frac{nV_{20}}{L} Z_1 \\ -\frac{nV_{20}}{L} Y_1 \\ -\frac{2n}{C_o}(I_{L1R}Z_1 - I_{L1I}Y_1) \end{bmatrix}$$

The state equation in (30) shows that we can reconstruct the state equation of the DAB converter through the coefficients of the Fourier series of the state variables v_2 and i_L . This paper only considers its 0th and 1st degree coefficients, considering more items will make the model more accurate, but it will bring computational complexity. In this section, the modeling process of the generalized state space average model of the DAB converter is described, which takes into account the equivalent resistance R_s , and the model has higher accuracy. Moreover, the model will not be affected by the transmission power modes, the model is universal and suitable for any transmission power modes.

V. SIMULATION AND EXPERIMENTAL VERIFICATION

To verify the correctness of the model proposed in section IV, the frequency response of the control signal to the output voltage is tested based on the simulation software PLECS and the experimental prototype, and the experimental prototype is shown in Figure 6. The electrical parameters of the simulation and experimental prototype are shown in Table I.

If the conditions of $d_1=0$ and $d_3=0$ are set, the model is simplified to SPS control; if the conditions of $d_1=0$ and $d_3 \neq 0$ ($d_1 \neq 0$ and $d_3=0$) are set, the model is simplified to EPS control, if the conditions of $d_1=d_3 \neq 0$ are set, the model is simplified to DPS control. In this paper, the control variables

d_1 , d_2 and d_3 are independent of each other when verifying the correctness of the model.

In Fig. 7, in addition to experimental results, the frequency response of the two models are compared: 1) the reduced-order model derived from transmit power in section III; 2) the proposed model in section IV.

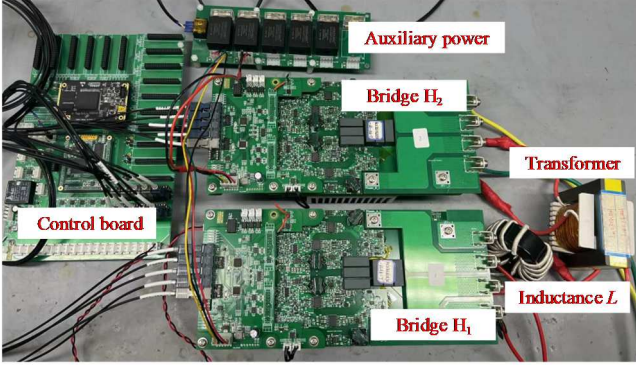


Fig. 6. Experimental prototype.

In this section, the frequency response results need to be measured at the static operating point of the converter. First, the control-to-output frequency response of the simulation results is obtained through the PLECS software. The frequency responses of the proposed model and reduced-order model are obtained by MATLAB/Simulink. In addition, the frequency response of the experimental results is obtained by measuring the output response by a single-point frequency perturbation [5], [6], and small perturbation frequencies from 4Hz to 2kHz. In the experiment, a sinusoidal small perturbation signal $d_{\sin}$ is added at the static operating point D of the control signal, then the actual control signal applied to the converter is $D + d_{\sin}$. Use the AC coupling function of the oscilloscope to measure the output voltage at this time. According to the small disturbance waveform and the output voltage waveform, the amplitude and phase information can be calculated out.

TABLE I. SIMULATION AND EXPERIMENTAL PARAMETERS

Parameters	Values
Switching frequency f_s	20kHz
HFT turns ratio n	1
Input voltage V_1	200V
Output capacitance C	2.35mH
Total inductance L	82.75μH
Total equivalent resistance R_s	0.086Ω
Load resistance R	20Ω
Phase shift d_1, d_2, d_3	0.1, 0.2, 0.1

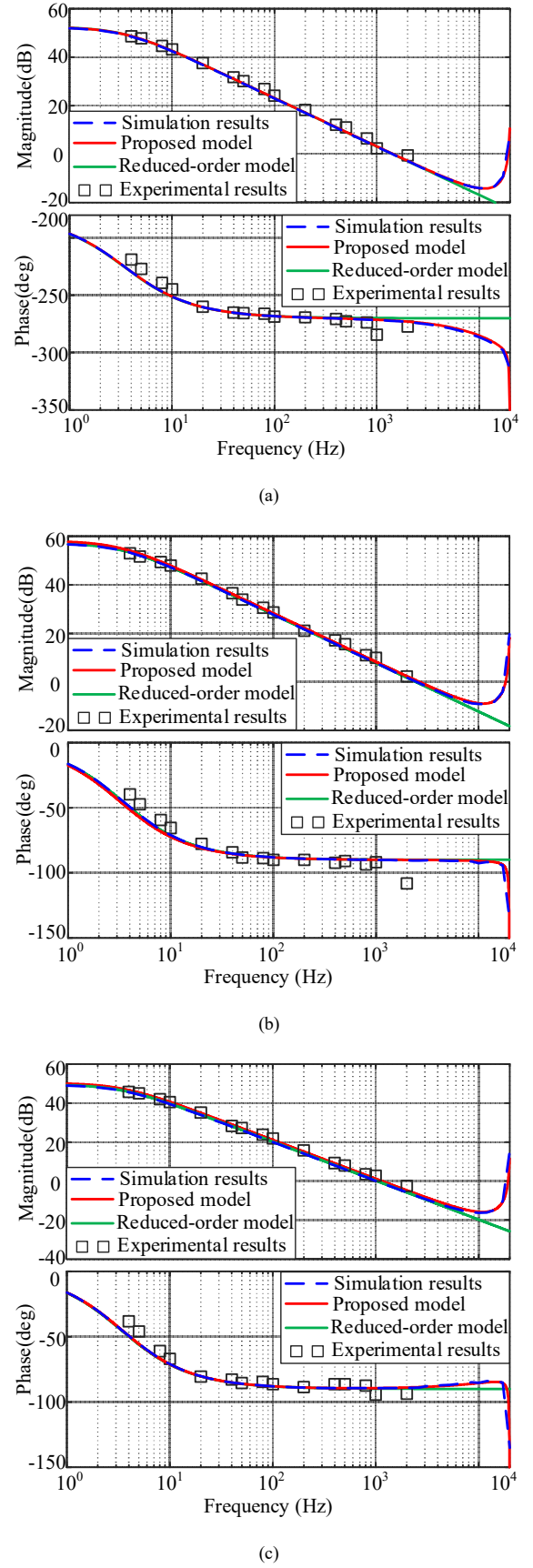


Fig. 7. Frequency response of control signal to output voltage (a)with d_1 , (b)with d_2 , (c)with d_3 .

Fig. 7 shows control-to-output frequency response. In the amplitude-frequency curve of Fig. 7, the reduced-order model differs from the simulation results at the switching frequency,

in particular, in the phase-frequency curve of Fig. 7(a), the reduced-order model differs greatly from the simulation results when the frequency is greater than 1 kHz, while the proposed model agrees well with the simulation results across the entire frequency range. The same situation also appears in the amplitude-frequency characteristic curves of Fig. 7(b) and Fig. 7(c). The reduced-order model is different from the simulation and experimental results after 1 kHz, while the proposed model has higher accuracy. In short, the proposed model is more matched with the experimental and simulation results than the reduced-order model.

In the experiment, due to the limitation of the switching frequency of the converter, after the frequency sweep experiment frequency exceeds 2 kHz, the oscilloscope cannot accurately obtain the output response waveform, and at this time, the waveform is severely distorted, and accurate amplitude and phase information cannot be obtained.

VI. CONCLUSION

In this paper, a generalized state space averaging model of DAB converter is established with TPS control. Compared with the reduced-order model, the proposed model has two advantages, one is that the model is more accurate, and the frequency response agrees with the simulation and experimental results over the entire frequency range. The other is that the model is suitable for TPS control, and the model is generic. The reduced-order model is affected by power transmission modes, and different transmission power modes will result in different large-signal models and small-signal models. While the proposed model does not need to be classified and considered according to the different power conditions. Through simulation and experiments, the correctness and accuracy of the proposed model are verified. The proposed model has important implications for the design and stability analysis of the DAB converter controller.

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